AI calcs (with example questions)

same colour = same number · yellow = worked out · red = careful

L2–3 SEARCH

Q. Frontier: **P (g = 120, h = 200)**, **Q (g = 90, h = 250)**, **R (g = 150, h = 160)**. Which node is expanded next by A*? By UCS? By greedy?

A* vs UCS vs Greedy (which node next)

P: g 120, h 200 Q: g 90, h 250 R: g 150, h 160

f = cost so far g + guess to goal h

A*: P 120+200 = 320, Q 90+250 = 340, R 150+160 = 310
→ smallest f → **R (310)**

UCS: smallest g only → **Q (90)** g = add up road costs so far

Greedy: smallest h only → **R (160)**

A* is optimal only if h never overestimates (admissible)

Q. Minimising energy. A move makes things worse by **3** at temperature **T = 2**. What is the probability the move is accepted?

Simulated annealing accept prob

ΔE = 3 (worse by 3) T = 2

$P = e^{-\Delta E / T}$

$= e^{-3/2} = e^{-1.5}$

$= \mathbf{0.2231}$

better move → always accept. high T = accepts more bad moves

Q. Fitness values **24**, **30**, **16**, **8**. Probability the first individual is picked by roulette wheel selection?

GA roulette wheel selection

fitness 24 30 16 8

$$P(\text{pick}) = \frac{\text{fitness}}{\text{sum of all fitness}}$$

sum = 24 + 30 + 16 + 8 = 78

$P = \frac{24}{78} = \mathbf{0.3077}$

Q. In the **8**-queens problem (one queen per column, move one queen within its column), how many successors does a state have?

N-queens successors

successors = n queens × (n − 1) other rows

$= 8 \times 7 = \mathbf{56}$

4-queens → $4 \times 3 = \mathbf{12}$

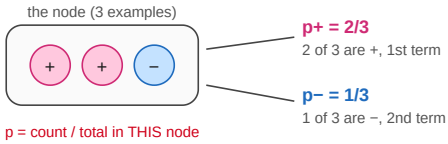
each queen can go to every row except the one it is on

L4 LEARNING FROM EXAMPLES

Q. A node has **2** positive and **1** negative examples. Find its entropy and its Gini index.

Entropy & Gini of a node

2 + 1 −



p = count / total in THIS node

$H = -p_+ \log_2 p_+ - p_- \log_2 p_-$

$= -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$

$= 0.39 + 0.5283 = \mathbf{0.9183}$

$Gini = 1 - (p_+^2 + p_-^2)$

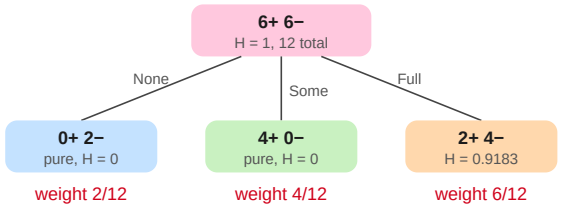
$= 1 - (\frac{4}{9} + \frac{1}{9}) = \mathbf{0.4444}$

pure node → 0. 50/50 → H = 1, Gini = 0.5

Q. Restaurant data **(6+, 6−)**. Patrons splits into None **(0+, 2−)**, Some **(4+, 0−)**, Full **(2+, 4−)**. Find IG(Patrons) and the Gini reduction.

Information gain (restaurant, Patrons)

root 6+ 6− None 0+ 2− Some 4+ 0− Full 2+ 4−



$$IG = H(\text{root}) - \Sigma \frac{\text{branch size}}{\text{total}} H(\text{branch})$$

H(root) = 1 None = 0, Some = 0 pure

H(Full) = H(2/6, 4/6) = 0.9183

$IG = 1 - [\frac{2}{12}(0) + \frac{4}{12}(0) + \frac{6}{12}(0.9183)]$

$= 1 - 0.4591 = \mathbf{0.5409}$

Gini: root 0.5, Full 0.4444 → $0.5 - \frac{6}{12}(0.4444) = \mathbf{0.2778}$

pick the attribute with the BIGGEST gain. every branch 50/50 (Type) → IG = 0

Q. Model **h(x) = 1 + 2x**. Data **(1, 4), (2, 4), (3, 8)**. Find the sum of squared errors.

Sum of squared errors

h(x) = 1 + 2x (1,4) (2,4) (3,8)

$SSE = \Sigma (\text{actual } y - \text{predicted } h(x))^2$

h(1) = 3, h(2) = 5, h(3) = 7

$= (4-3)^2 + (4-5)^2 + (8-7)^2$

$= 1 + 1 + 1 = \mathbf{3}$

MSE → divide by n (= 1). some slides use ½ SSE

Q. Classify **(2, 3)** with k = 3 (Euclidean). Points: (1,1) A, (3,4) B, (4,3) B, (0,5) A, (6,6) A.

KNN classify (k = 3)

query (2,3) **A (1,1)** **B (3,4)** **B (4,3)** **A (0,5)** **A (6,6)**

$d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$

A(1,1): $\sqrt{1+4} = 2.24$ **B(3,4)**: $\sqrt{1+1} = 1.41$

B(4,3): $\sqrt{4+0} = 2$ **A(0,5)**: $\sqrt{4+4} = 2.83$

A(6,6): $\sqrt{16+9} = 5$

3 closest: B, B, A → **B**

no need to square root if you only compare

Q. **TP = 40**, **FP = 10**, **FN = 20**, **TN = 30**. Find accuracy, precision, recall and F1.

Precision, recall, F1

TP 40 **FP 10** **FN 20** **TN 30**

$$\text{acc} = \frac{TP + TN}{all} \quad P = \frac{TP}{TP + FP} \quad R = \frac{TP}{TP + FN}$$

$$\text{acc} = \frac{40 + 30}{100} = \mathbf{0.7}$$

$$P = \frac{40}{40 + 10} = 0.8 \quad R = \frac{40}{40 + 20} = 0.667$$

$$F1 = \frac{2PR}{P + R}$$

$$= \frac{2(0.8)(0.667)}{0.8 + 0.667} = \mathbf{0.7273}$$

P → of the ones I said yes. R → of the real yes

L5 NEURAL NETS

Q. Step perceptron, x = **(1, 1)**, w = **(0.3, 0.2)**, bias **−0.4**. Target **0**, learning rate **0.1**. Find the output and the new w1.

Perceptron output + update

x = (1, 1) **w = (0.3, 0.2)** **b = −0.4** **target y = 0** **η = 0.1**

$\text{out} = \text{step}(\overset{}{w}\cdot\overset{}{x} + \overset{}{b})$

$= 0.3(1) + 0.2(1) - 0.4 = 0.1 \geq 0$ → out = 1

$$w_1 \text{ new} = \overset{0.3}{w1} + \overset{0.1}{\eta}(\overset{0}{y} - \text{out})\overset{1}{x1}$$

$= 0.3 + 0.1(0 - 1)(1) = \mathbf{0.2}$

y = target, out is 0 or 1 (not 0.1). right answer → no change

---

Q. x1 = **0**, x2 = **2**. Hidden unit 3 is ReLU with w1,3 = **−0.2**, w2,3 = **−0.4**. Output unit 5 is sigmoid with w3,5 = **1.3**. No bias. Find a3 and the output.

Forward pass (ReLU then sigmoid)

x1 = 0 **x2 = 2** **w1,3 = −0.2** **w2,3 = −0.4** **w3,5 = 1.3**

$$z_3 = \overset{0}{x1}\overset{-0.2}{w1,3} + \overset{2}{x2}\overset{-0.4}{w2,3}$$

$= 0(-0.2) + 2(-0.4) = -0.8$

$a_3 = \text{ReLU}(-0.8) = 0$

$z_5 = 0 \times 1.3 = 0$ → $\sigma(0) = \frac{1}{1 + e^0} = \mathbf{0.5}$

ReLU: negative → 0. $\sigma(z) = 1 / (1 + e^{-z})$

Q. Find sigmoid(0), ReLU(**−3**), and the loss $(y - \hat{y})^2$ when target **y = 1** and output **$\hat{y}$ = 0.8**.

Activations & loss (quick ones)

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad \text{ReLU}(z) = \max(0, z) \quad \tanh(0) = 0$$

$$\sigma(0) = \frac{1}{1 + 1} = \mathbf{0.5} \quad \text{ReLU}(-3) = \mathbf{0}$$

$$\text{loss} = (\overset{1}{y} - \overset{0.8}{\hat{y}})^2$$

$= (1 - 0.8)^2 = 0.2^2 = \mathbf{0.04}$

square AFTER subtracting

L6 REINFORCEMENT LEARNING

Q. Rewards **−1, −1, +10** with γ = **0.5**. Find the discounted return.

Discounted return

rewards −1, −1, +10 **γ = 0.5**

$$G = r_1 + \overset{0.5}{\gamma} r_2 + \gamma^2 r_3$$

$= -1 + 0.5(-1) + 0.25(10)$

$= -1 - 0.5 + 2.5 = \mathbf{1}$

first reward is NOT discounted

Q. Grid world: intended move **0.8**, each side **0.1**. R = **−0.04**, γ = **0.5**. Utilities: intended square **0.39**, sides **0.25** and **0.04**. Find Q(s, a).

Q-value with slippery moves

0.8 intended **0.1 / 0.1 sideways** **R = −0.04** **γ = 0.5**

V: 0.39, 0.25, 0.04

$$Q = \Sigma\, P(\overset{-0.04}{R} + \overset{0.5}{\gamma}\, V(s'))$$

$= 0.8(-0.04 + 0.5 \times 0.39) + 0.1(-0.04 + 0.5 \times 0.25) + 0.1(-0.04 + 0.5 \times 0.04)$

$= 0.8(0.155) + 0.1(0.085) + 0.1(-0.02)$

$= 0.124 + 0.0085 - 0.002 = \mathbf{0.1305}$

bump into wall → you stay, use your own V

Q. Up mostly reaches A = **0.32**, Right mostly reaches B = **0.44**, the other side is C = **0.21**. Moves go 0.8 intended, 0.1 to each side. Which action does the policy pick?

Pick the best action (policy)

A = 0.32 **B = 0.44** **C = 0.21**

value(move) = 0.8 × target + 0.1 × left side + 0.1 × right side

Up = $0.8(0.32) + 0.1(0.44) + 0.1(0.21) = 0.321$

Right = $0.8(0.44) + 0.1(0.32) + 0.1(0.21) = 0.405$

biggest → **Right**

policy = argmax, so give the ACTION not the number

Q. TD: V(s) = 0.52, V(s′) = 0.21, r = 0, α = 0.2, γ = 0.9. Find the new V(s). Then Q-learning with Q(s,a) = 0.97 and max Q(s′,a′) = 1.63.

TD & Q-learning

V(s) = 0.52 V(s′) = 0.21 α = 0.2 γ = 0.9 r = 0

$V(s) \leftarrow \overset{0.52}{V(s)} + \overset{0.2}{\alpha}\left(\overset{0}{r} + \overset{0.9}{\gamma}\,\overset{0.21}{V(s')} - \overset{0.52}{V(s)}\right)$

= 0.52 + 0.2(0 + 0.9×0.21 − 0.52)

= 0.52 + 0.2(−0.331) = **0.4538**

Q(s,a) = 0.97 max Q(s′) = 1.63

$Q \leftarrow Q + \alpha(r + \gamma \max Q(s',a') - Q)$

= 0.97 + 0.2(0 + 0.9×1.63 − 0.97)

= 0.97 + 0.2(0.497) = **1.0694**

Q-learning uses the MAX of next state. SARSA uses the action taken. U(s) = V(s)

Q. Route A: 80% chance U = +10, 20% chance U = −10. Route B: 100% U = +7. Which does MEU choose?

Expected utility (MEU)

$EU = \Sigma\, P \times U$

A = 0.8(10) + 0.2(−10) = 8 − 2 = 6

B = 1(7) = 7

7 > 6 → **B**

MEU = biggest expected value, even if A can win more

Q. Two trials give reward-to-go 0.84 and 0.92 for state s. Estimate U(s).

Direct utility estimate

U(s) = average of reward-to-go seen from s

$= \frac{0.84 + 0.92}{2} = \mathbf{0.88}$

reward-to-go = sum of rewards from that state to the end

Q. Q(s,·): Up 0.4, Down 0.7, Left −0.2, Right 0.1. Find V(s) and the greedy action. With ε = 0.2 over 4 actions, find P(greedy) and P(other).

Q(s,a) from a Q-table + ε-greedy

Up 0.4 Down 0.7 Left −0.2 Right 0.1 ε = 0.2

$V(s) = \max_a Q(s,a)$ $\pi(s) = \text{argmax}_a\, Q(s,a)$

max(0.4, 0.7, −0.2, 0.1) = **0.7** → **Down**

$P(\text{greedy}) = (1 - \overset{0.2}{\varepsilon}) + \frac{\overset{0.2}{\varepsilon}}{|A|}$

$= 0.8 + \frac{0.2}{4} = 0.8 + 0.05 = \mathbf{0.85}$ P(other) = **0.05**

max = value, argmax = action. random pick excludes greedy → 1 − ε and ε/3

Q. Q(s,a) = 0.5, r = 1, γ = 0.9, α = 0.1. The agent takes a′ with Q = 0.2, but the best Q in s′ is 0.8. New Q with SARSA? With Q-learning?

SARSA vs Q-learning

Q = 0.5 r = 1 γ = 0.9 α = 0.1 taken 0.2, best 0.8

$Q \leftarrow \overset{0.5}{Q} + \overset{0.1}{\alpha}\left(\overset{1}{r} + \overset{0.9}{\gamma}\,\overset{}{Q(\text{next})} - \overset{0.5}{Q}\right)$

SARSA (next = action TAKEN, 0.2):

= 0.5 + 0.1(1 + 0.9×0.2 − 0.5) = 0.5 + 0.1(0.68) = **0.568**

Q-learning (next = MAX, 0.8):

= 0.5 + 0.1(1 + 0.9×0.8 − 0.5) = 0.5 + 0.1(1.22) = **0.622**

SARSA = on-policy. Q-learning = off-policy. both model-free

Q. All Q start at 0, α = 0.5, γ = 1. Each episode: A →(r 0) B →(r +10) Goal (terminal). Q(A,right) after episodes 1 and 2?

Q-learning over episodes

α = 0.5 γ = 1 end reward 10

$Q \leftarrow Q + \overset{0.5}{\alpha}\left(r + \overset{1}{\gamma} \max Q(\text{next}) - Q\right)$ terminal → max Q(next) = 0

ep 1, A: 0 + 0.5(0 + 1×0 − 0) = 0

ep 1, B: 0 + 0.5(10 − 0) = 5

ep 2, A: 0 + 0.5(0 + 1×5 − 0) = **2.5**

update in the order visited. the reward reaches A one episode later

---

Q. Deterministic: r = −1, Q(s′,·) = 2, 5, 3, γ = 0.9. Q(s,a)? Then: Q = 2, step to TERMINAL with r = 5, α = 0.5. New Q?

Q(s,a): deterministic or terminal

$Q(s,a) = \overset{-1}{r} + \overset{0.9}{\gamma} \max \overset{5}{Q(s',a')}$

= −1 + 0.9×5 = **3.5**

$\text{terminal: target} = \overset{5}{r}\ \text{only}$

Q = 2 + 0.5(5 − 2) = **3.5**

no γ × anything after a terminal state

L7 CNN

Q. Input 7×7×16, 32 filters of size 5×5, stride 2, padding 2, no bias. Find the output size and the number of parameters.

Conv output size & parameter count

input 7×7×16 32 filters filter 5×5 stride 2 padding 2

$\text{output} = \frac{\overset{7}{\text{input}} + 2\,\overset{2}{\text{padding}} - \overset{5}{\text{filter size}}}{\underset{2}{\text{stride}}} + 1$

$= \frac{7 + 4 - 5}{2} + 1 = 3 + 1 = 4 \rightarrow \mathbf{4 \times 4 \times 32}$

$\text{params} = (\overset{5}{\text{filter}} \times \overset{5}{\text{filter}}) \times \overset{16}{\text{channel}} \times \overset{32}{\text{conv layer}}$

= 5×5×16×32 = **12 800**

+ bias → + 1 per filter = 12 800 + 32 = 12 832

channel = depth of the INPUT (RGB = 3). depth of output = no. of filters

Q. Apply the kernel to the 3×3 patch below (no bias). What is the output value?

Conv, one output value

3	1	0
2	0	1
1	4	1

$*$

1	0	−1
1	0	−1
1	0	−1

value = sum of (patch × kernel) + bias

= (3−0) + (2−1) + (1−1) + 0

= 3 + 1 + 0 = **4**

middle column × 0 drops out. slide by stride for the next one

---

Q. 2×2 pooling with stride 2 on the 4×4 map below. Find the max-pool output, and the average for the bottom-right block.

Max & average pooling (2×2, stride 2)

5	9	2	3
1	9	3	0
9	6	5	6
4	6	6	0

max: [5 9 1 9] = 9 [2 3 3 0] = 3

[9 6 4 6] = 9 [5 6 6 0] = **6**

avg (bottom right): $\frac{5+6+6+0}{4} = \mathbf{4.25}$

split into blocks, one number per block → 4×4 becomes 2×2

Q. Layers: conv 3×3, conv 3×3, max pool 2×2 stride 2, conv 3×3 (convs stride 1). What is the receptive field of one output?

Receptive field

conv 3 conv 3 pool 2, stride 2 conv 3

r = r + (k − 1) × jump jump ×= stride

start r = 1, jump = 1

conv 3: 1 + 2(1) = 3

conv 3: 3 + 2(1) = 5

pool 2 s2: 5 + 1(1) = 6 → jump = 2

conv 3: 6 + 2(2) = **10**

jump only changes AFTER the strided layer

Q. Box A (0,0)–(6,4), box B (2,0)–(8,4), same class, A has the higher score. Find IoU. With NMS threshold 0.5, is B kept?

IoU & NMS

box A (0,0)–(6,4) box B (2,0)–(8,4) threshold 0.5

$$\text{IoU} = \frac{\text{overlap}}{\text{area A + area B − overlap}}$$

overlap = (6 − 2)(4) = 16, each box = 24

$\text{IoU} = \frac{16}{24 + 24 - 16} = \frac{16}{32} = \mathbf{0.5}$

≥ 0.5 → same object → drop the lower score box

NMS: keep top score, delete boxes overlapping it, repeat

L8 RNN, ATTENTION, NLP

Q. Simple RNN, tanh. x = 0, input weight 0.5, previous hidden state 0.1, recurrent weight 0.75, bias 0.1. Find h_t.

Simple RNN hidden state update

x = 0 w = 0.5 h_{t-1} = 0.1 u = 0.75 b = 0.1

$h_t = \tanh(\overset{0.5\,0}{w\,x} + \overset{0.75\ 0.1}{u\ h_{t-1}} + \overset{0.1}{b})$

= tanh(0.5(0) + 0.75(0.1) + 0.1)

= tanh(0.175)

= **0.1732**

u = recurrent weight (on old h). small input → tanh(z) ≈ z

Q. Attention scores 2 and 1. Find the softmax attention weights (4 d.p.).

Softmax attention weights

score 1 = 2 score 2 = 1

$$\alpha_1 = \frac{e^{\text{score 1}}}{e^{\text{score 1}} + e^{\text{score 2}}}$$

$= \frac{e^2}{e^2 + e^1} = \frac{7.3891}{7.3891 + 2.7183}$

$= \frac{7.3891}{10.1073} = \mathbf{0.7311}$

$\alpha_2 = 1 - 0.7311 = 0.2689$

equal scores → equal weights (2, 2 → 0.5 each)

Q. Attention weights 0.7 and 0.3, value vectors [0, 2] and [2, 0]. Find the output.

Attention output (weighted sum)

α = 0.7, 0.3 v1 = [0, 2] v2 = [2, 0]

$\text{out} = \overset{0.7}{\alpha 1}\,v1 + \overset{0.3}{\alpha 2}\,v2$

= 0.7[0, 2] + 0.3[2, 0]

= [0, 1.4] + [0.6, 0] = **[0.6, 1.4]**

weights must add to 1 (from softmax)

Q. "the cat" appears 3 times and "the" appears 5 times. Find P(cat | the).

Bigram probability

count("the cat") = 3 count("the") = 5

$$P(\text{cat} \mid \text{the}) = \frac{\overset{3}{\text{count(the cat)}}}{\underset{5}{\text{count(the)}}}$$

$= \frac{3}{5} = \mathbf{0.6}$

sentence prob → multiply each bigram

L10 LOGIC

Q. With P = True, Q = False, R = False, is ¬P ∨ Q ∧ R ⇒ R true or false?

Logic, evaluate a sentence

P = T Q = F R = F

¬P ∨ Q ∧ R ⇒ R

order: ¬ first, then ∧, then ∨, then ⇒

(¬T) ∨ (F ∧ F) ⇒ F

= (F ∨ F) ⇒ F

= F ⇒ F = **True**

⇒ is only false when T ⇒ F

L11 PROBABILITY & BAYES NETS

Q. $P(A) = 0.3$, $P(B) = 0.5$, $P(A \wedge B) = 0.15$. Find $P(A \vee B)$. Then, if $P(A \wedge B) = 0.12$ and $P(B) = 0.4$, find $P(A \mid B)$.

Probability rules

$P(A) = 0.3$ $P(B) = 0.5$ $P(A \wedge B) = 0.15$

$$P(A \vee B) = \overset{0.3}{P(A)} + \overset{0.5}{P(B)} - \overset{0.15}{P(A \wedge B)}$$

$= 0.3 + 0.5 - 0.15 = \underline{\underline{0.65}}$

$P(A \wedge B) = 0.12$ $P(B) = 0.4$

$$P(A|B) = \frac{\overset{0.12}{P(A \wedge B)}}{\underset{0.4}{P(B)}}$$

$= \frac{0.12}{0.4} = \underline{\underline{0.3}}$

$P(\neg A) \rightarrow 1 - P(A)$

Q. Using the joint table below, find $P(a \mid b)$.

Joint table, P(a|b)

	b	¬b
a	0.132	0.418
¬a	0.25	0.2

$$P(a|b) = \frac{P(a, b)}{P(b)}$$

$P(b) = P(a,b) + P(\neg a,b) = 0.132 + 0.25 = 0.382$ ← a cancels out

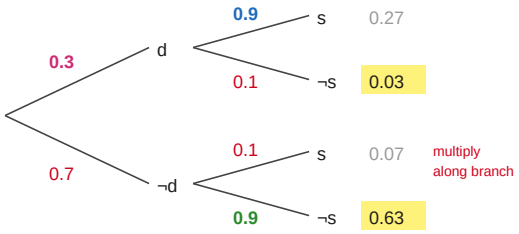
$P(a|b) = \frac{0.132}{0.382} = \underline{\underline{0.3455}}$

to get P(b) add the whole b column

Q. 30% of patients have disease d. The test is positive for 90% of sick patients and negative for 90% of healthy ones. A patient tests negative. Find $P(d \mid \neg s)$.

Bayes with a negative result (tree)

$P(d) = 0.3$ $P(s|d) = 0.9$ $P(\neg s|\neg d) = 0.9$



$$P(d|\neg s) = \frac{P(\neg s|d)\,P(d)}{P(\neg s)}$$

$P(\neg s) = (1-0.9)(0.3) + 0.9(1-0.3)$

$= 0.03 + 0.63 = 0.66$

$P(d|\neg s) = \frac{0.03}{0.66} = \underline{\underline{0.0455}}$

negative result → use the ¬s branches. 1 − P to flip

Q. 1% of people have disease D. $P(+ \mid D) = 0.9$, $P(+ \mid \neg D) = 0.05$. Someone tests positive. Find $P(D \mid +)$.

Disease test (Bayes)

$P(D) = 0.01$ $P(+|D) = 0.9$ $P(+|\neg D) = 0.05$

$$P(D|+) = \frac{\overset{0.9}{P(+|D)}\ \overset{0.01}{P(D)}}{P(+)}$$

$P(+) = 0.9 \times 0.01 + 0.05 \times (1-0.01)$

$= 0.009 + 0.0495 = 0.0585$

$P(D|+) = \frac{0.009}{0.0585} = \underline{\underline{0.1538}}$

false alarm rate uses ¬D, which is 0.99 of people

Q. Machines M1, M2, M3 make 50%, 30%, 20% of parts, with defect rates 2%, 3%, 5%. Find P(defective) and P(M3 | defective).

Which machine made the defect

M1 50%, 2% bad M2 30%, 3% bad M3 20%, 5% bad

P(D) = sum of P(M) P(D|M)

$= 0.5 \times 0.02 + 0.3 \times 0.03 + 0.2 \times 0.05$

$= 0.01 + 0.009 + 0.01 = 0.029$

$P(M3|D) = \frac{0.01}{0.029} = \underline{\underline{0.3448}}$

Q. Burglary network. Find P(j, ¬m, ¬a, ¬b, ¬e): John calls, Mary does not, no alarm, no burglary, no earthquake.

Bayes net, one full assignment

$P(j, \neg m, \neg a, \neg b, \neg e) = P(\neg b)\,P(\neg e)\,P(\neg a|\neg b,\neg e)\,P(j|\neg a)\,P(\neg m|\neg a)$

each node given its parents only

$= 0.999 \times 0.998 \times 0.999 \times 0.05 \times 0.99$

$= \underline{\underline{0.0493}}$

table gives P(a|...) → P(¬a|...) = 1 − that

Q. Burglary network with the CPTs below. John and Mary both call. Find P(Burglary | j, m) by enumeration (4 d.p.).

Bayes net enumeration P(B | j, m)

P(b) = 0.01 P(e) = 0.02 P(a|b,e) .95 .94 .3 .05 P(j|a) .9 / .05
P(m|a) .6 / .02

$$P(b|j,m) = \alpha \overset{0.01}{P(b)} \Sigma_e \overset{0.02}{P(e)} \Sigma_a P(a|b,e)\ P(j|a)\ P(m|a)$$

1) j, m part for a and ¬a 2) inside bracket for e and ¬e 3) S(b), S(¬b) 4) divide

P(j|a) P(m|a)

= .9 × .6 = .54

P(j|¬a) P(m|¬a)

= .05 × .02 = .001

S(b)

[b, e] = P(a|b,e)(.54) + P(¬a|b,e)(.001)

= .95(.54) + .05(.001) = 0.51305 ¬a = 1 − .95

[b, ¬e] = P(a|b,¬e)(.54) + P(¬a|b,¬e)(.001)

= .94(.54) + .06(.001) = 0.50766

S(b) = P(b) [P(e)[b, e] + P(¬e)[b, ¬e]]

= .01 [.02(0.51305) + .98(0.50766)]
= .01 [0.010261 + 0.497507] = 0.005078

S(¬b)

[¬b, e] = P(a|¬b,e)(.54) + P(¬a|¬b,e)(.001)

= .3(.54) + .7(.001) = 0.1627

[¬b, ¬e] = P(a|¬b,¬e)(.54) + P(¬a|¬b,¬e)(.001)

= .05(.54) + .95(.001) = 0.02795

S(¬b) = P(¬b) [P(e)[¬b, e] + P(¬e)[¬b, ¬e]]

= .99 [.02(0.1627) + .98(0.02795)] P(¬b) = 1 − .01
= .99 [0.003254 + 0.027391] = 0.030339

$$P(b|j,m) = \frac{S(b)}{S(b) + S(\neg b)}$$

$$= \frac{0.005078}{0.005078 + 0.030339} = \mathbf{0.1434}$$

do b and ¬b the same way, then divide by the total (that is α)

Q. "great" appears 8 times in positive reviews (25 words total) and 0 times in negative reviews (14 words). Vocab size 10, equal priors. With Laplace smoothing, find P(+ | "great").

Naive Bayes (text)

"great" in + : 8 words in + : 25 vocab : 10 "great" in − : 0
words in − : 14 P(+) = P(−) = 0.5

$$P(\text{great}|+) = \frac{\overset{8}{\text{word appears}} + 1}{\underset{25}{\text{words in +}} + \underset{10}{\text{vocab}}}$$

$$= \frac{8 + 1}{25 + 10} = \frac{9}{35} = 0.2571$$

$$P(\text{great}|-) = \frac{0 + 1}{14 + 10} = \frac{1}{24} = 0.04167$$

+ : 0.5 × 0.2571 = 0.12857 − : 0.5 × 0.04167 = 0.02083

$$P(+|\text{text}) = \frac{0.12857}{0.12857 + 0.02083} = \mathbf{0.8606}$$

+1 so a zero count does not kill the whole product

Q. Using the joint table below, find P(x1), P(z1) and P(x1 | z1).

Marginalise from a full joint table

X	Y	Z	P
x1	y1	z1	.06
x1	y1	z2	.09
x1	y2	z1	.14
x1	y2	z2	.11
x2	y1	z1	.12
x2	y1	z2	.18
x2	y2	z1	.20
x2	y2	z2	.10

P(x1) = add every row with x1 (sum out Y, Z)

P(x1) = .06 + .09 + .14 + .11 = **0.40**
P(z1) = .06 + .14 + .12 + .20 = 0.52
P(x1, z1) = .06 + .14 = 0.20

$$P(x1 | z1) = \frac{0.20}{0.52} = \mathbf{0.3846}$$

given part goes on the bottom

Q. How many rows does a full joint table for 5 binary variables have? How many models for 7 propositional symbols?

Counting rows / models

rows = 2^n (n = number of binary variables)

2^5 = **32** 2^7 = **128**

each binary variable doubles it. 3 values each → 3^n

FILL-IN-THE-BLANK ONE-LINERS

Formulas they blank out

Greedy f(n) = **h(n)** UCS f(n) = **g(n)** A* f(n) = **g(n) + h(n)**
BFS = **FIFO queue** DFS = **LIFO stack**
IG = H(parent) − **weighted avg H(children)** pure node H = **0**, 50/50 H = **1**
σ(0) = **0.5** ReLU(neg) = **0** tanh(0) = **0**
U(s) = R(s) + γ **$\max_a \Sigma P(s'|s,a) U(s')$** (Bellman)
VAE reparam: z = μ + **σ** · ε
P ⇒ Q ≡ **¬P ∨ Q** P ⇔ Q ≡ **(P ⇒ Q) ∧ (Q ⇒ P)**
¬(A ∧ B) ≡ **¬A ∨ ¬B** ¬(A ∨ B) ≡ **¬A ∧ ¬B**
P ⇒ Q ≡ **¬Q ⇒ ¬P** (contrapositive)
marginalise = **sum** over the other variables joint rows = **2^n**
P(A|B) = **P(A ∧ B) / P(B)** normalise: α = **1 / (s(b) + s(¬b))**
category reasoning: perceive → **categorise** → predict
the bold part is what usually gets blanked